Intelligent Medical Report Analysis System

Project Domain: DL / GenAI / NLP

DSAI LAB Group 2

Indian Institute Of Technology, Madras

TEAM

01

Shivendra Patel
21f2001310@ds.study.iitm.ac.in
S.P.

02

Samta Ranka
23f1001316@ds.study.iitm.ac.in
S.R.

03

Rajat Srivastava
22f3003195@ds.study.iitm.ac.in

04

Ritwik Trivedi
22f1000120@ds.study.iitm.ac.in

05

Bryan David Robinson
22f3002277@ds.study.iitm.ac.in

TRAINING DATA SOURCES

MTSamples Dataset
Source: Scraped from mtsamples.com, available on Kaggle
Format: CSV with 6 columns: description, medical_specialty, sample_name, transcription, keywords
Size: 4,999 total records, 2,416 lab-related records identified through keyword filtering (48.3% of total)
License: Open access (HIPAA-compliant, de-identified)

MIMIC-III Dataset
Source: MIT Lab for Computational Physiology, available via PhysioNet
Version: v1.4 (released September 2016)
Access: Credentialed access required (CITI training, Data Use Agreement)
Ethics: De-identified, IRB-approved, HIPAA-compliant
Sample from Kaggle: Available on Kaggle MIMIC-III KAGGLE LINK
Format: CSV files including LABEVENTS, D_LABITEMS, ADMISSIONS, PATIENTS
Size (as loaded from Kaggle sample): LABEVENTS: 76,074 rowsD_LABITEMS: 753 rows (laboratory test definitions)ADMISSIONS: 129 rows (patient admission records)
Patients: 100 unique patients

MTSamples (Secondary)	Lab-related transcriptions containing explanatory text about what test results mean (e.g., "The elevated creatinine level is consistent with acute renal failure") .	Used to extract additional explanatio n-text pairs.

TESTING DATA SOURCES

Sample Medical Report PDFs/Images
Source: Curated collection from publicly available diagnostic lab websites and sample report repositories
Format: PDF and image files (PNG, JPG) containing structured laboratory reports
Sample Sources:
- Labsmart Sample Reports (https://www.labsmartlis.com/sample-reports)
- Drlogy Pathology Lab Report Templates (https://www.drlogy.com/pathology-lab-software/report-format)
- Additional diagnostic lab sample reports from Google Images and other publicly available sources

Size: 50-100 sample reports covering diverse layouts, test panels, and formats
Purpose: To evaluate the complete pipeline (OCR → NER → Abnormality Detection → Dashboard Generation) on realistic report formats that mimic what end-users would upload

Aspect	Training Data (MTSamples/MIMIC-III)	Testing Data (Sample PDFs/Images)
Format	Plain text/CSV	PDF/Image
Content	Clinical narratives and structured lab notes	Complete lab reports with layout, formatting, branding
Purpose	Train NER model to recognize entities in text	Test end-to-end pipeline on realistic documents
Privacy	De-identified; publicly available	Publicly available samples; no real patient data

DATASET DESCRIPTION

Dataset	Key Features	Statistics
MTSamples	Clinical notes (`transcription`), specialty, description, keywords	**4,999 records**, **2,416 lab-related (48.3%)**
MIMIC-III	Lab events, test definitions, admissions	**(76,074 lab events**, **753 lab tests - Full Dataset access required)**, (**129 admissions**, **65,280 valid numerical results**, **18 common lab tests**, **100 synthetic reports** from **100 patients - open access)**
Testing Dataset	PDF/Image lab reports with patient details, test tables, metadata	Multiple lab templates, variations in QR codes, TAT display, page numbering, abnormal-result highlighting, and test order

DATASET PREPROCESSING

Data Sources

•**MTSamples (4,999 reports)** ⟶ Keyword filtering for lab-related terms ⟶ **2,416 lab reports**

•**MIMIC-III** (LABEVENTS, D_LABITEMS, ADMISSIONS) ⟶ Numerical lab value filtering ⟶ **65,280 valid events** ⟶ **100 synthetic patient reports**

Combined Dataset

•Total Reports: **2,512**

- MTSamples: **2,412 (96%)**
- MIMIC-III: **100 (4%)**

•Exported in **CSV** and **JSONL** formats with source labels

Entity Annotation

•Rule-based annotation using regex patterns:

- Test Names (**31 patterns**)
- Units (**13 patterns**)
- Values & Reference Ranges

•Total Entities Identified: **52,768**

Dataset Split (Seed = 42)

•Train: **1,759 (70%)**

•Validation: **376 (15%)**

•Test: **377 (15%)**

•Stratified to preserve source distribution

Testing Dataset Quality

•Real lab report templates from diagnostic providers

•Standardized format with test names, values, units, and reference ranges

•Multiple sources used to improve diversity

DATASET STATISTICS

Text Length Statistics

Split	Mean	Min	Max	Std Dev
Training	3,534 chars	318 chars	18,425 chars	2,132
Validation	3,550 chars	318 chars	15,216 chars	2,223
Test	3,552 chars	347 chars	12,852 chars	1,974

Source Distribution

Source	Count	Percentage
MTSamples	2,412	96.0%
MIMIC-III	100	4.0%
Total	**2,512**	**100%**

Entity Annotation Results

Split	Reports	Total Entities
Training	1,759	52,768
Validation	376	(processed)
Test	377	(processed)

DATA ADEQUACY ASSESSMENT

Training Data Adequacy

Requirement	MTSamples	MIMIC-III	Combined
Clinical text with lab values	Yes	Yes (synthetic reports)	Yes
Lab test names	Varies	Structured	Comprehensive
Numeric values	Varies	Structured	Good coverage
Reference ranges	Limited	Included	Available
Report length	Average 3,500 chars	Average 3,500 chars	Consistent

Testing Data Adequacy

Requirement	Sample Reports
Realistic report layout	Yes (actual lab templates)
Diverse formats	Yes (multiple sources, versions)
OCR testability	Yes (PDF/Image format)
All required fields	Yes (test, value, unit, range)

FUTURE DATASET EXTENSION PLANS

Datasets Currently Being Sought:

- Complete MIMIC-III (Full PhysioNet version): Credentialed access in progress
- MIMIC-IV: Updated version with more recent data
- eICU Collaborative Research Database: Multi-center ICU lab results

Justification for Future Data:

- MIMIC-III Complete: Will provide larger patient cohort for better model generalization
- MIMIC-IV: More recent data for improved clinical relevance
- eICU: Different patient population (ICU focus) for broader validation

Synthea Dataset:

- Status: Downloaded but not used for training
- Reason: Contains synthetic data; real patient data (MIMIC-III, MTSamples) is preferred for training a clinically accurate model

Testing Data:

- Status: Actively being collected from publicly available sources
- Purpose: Validate complete pipeline on realistic document layouts

AUGMENTATION STRATEGY

Training

- Combine **MTSamples** and **MIMIC-III** to improve lab entity coverage.
- Add **MIMIC-IV, eICU, NHANES, and Synthea** for greater data diversity.
- Continuously expand the **reference range dictionary** for additional lab tests.

Testing

- Collect more sample reports from diagnostic lab websites.
- Generate synthetic reports to cover rare formats.
- Create abnormal-value variations to improve robustness.

DATALEKAGE PREVENTION

- **Patient-Level Splitting:** MIMIC-III reports grouped by patient ID (100 unique patients).
- **No Data Overlap:** Each patient's records appear in only one dataset split.
- **Stratified Splitting:** Source distribution maintained across training, validation, and test sets.
- **Independent Test Set:** Testing samples are completely separate from training data.
- **Different Data Sources:** Training data from MTSamples/MIMIC-III; testing data from sample report PDFs.
- **Reproducibility:** Fixed random seed (**42**) used for dataset splitting.
- **Source Tracking:** Dataset source labels retained for traceability and analysis.

Key Outcome: Prevents information leakage and ensures reliable model evaluation.

REFERENCE RANGES

Reference ranges defined for **29 common laboratory tests** used in the deterministic abnormality detection module.

Categories Covered:

- **Hematology:** Hemoglobin, WBC, RBC, HCT, Platelets
- **Kidney Function:** Creatinine, BUN, Uric Acid
- **Electrolytes & Minerals:** Sodium, Potassium, Chloride, Calcium, Magnesium, Phosphorus
- **Liver Function:** AST, ALT, Bilirubin, Albumin, Alkaline Phosphatase, Total Protein, Globulin
- **Coagulation:** INR, PT, PTT
- **Endocrine & Metabolic:** Glucose, HbA1c, TSH, T4
- **Iron Studies:** Iron, Ferritin

Automated Entity Annotation Engine

Algorithmic Labeling:

Because manual labeling is highly time-consuming, we implemented a custom, rule-based `MedicalEntityAnnotator` class.

Pattern Matching Logic

Utilized precise regular expression parsing structures to locate and tag tokens automatically across four custom clinical entities:

- **TEST Names:** Mapped using 31 distinct clinical pattern structures.
- **UNIT Categories:** Isolated using 13 distinct measurement unit patterns.
- **VALUE Fields:** Captured purely numeric data blocks.
- **REFERENCE_RANGE Bounds:** Captured lab limits.

Total Yield

Successfully extracted and labeled **52,768 total medical entities** for model training.

Stratified Splitting & Data Leakage Prevention

Strict Evaluation Safeguards

- 70 / 15 / 15 Stratified Split: Split using random seed 42 to ensure exact technical reproducibility while preserving source distribution across all sets.
- Patient-Level Isolation: MIMIC-III data is grouped entirely by patient ID (subject_id) before splitting. All records for a given patient exist inside a single split, ensuring no data crosses between training and validation.
- Domain Separation: Training text (narratives/sub-samples) and testing files (PDF/Image layouts) are sourced from entirely separate places.
- Algorithmic Cross-Check: Scripts execute a string hash intersection verification pass to guarantee 0 rows of unexpected data overlap.

THANKYOU